

ZANJAN, Iran, Islamic Republic of

WMO: 407290

Lat: 36.6598N

Lon: 48.5218E

Elev: 5456

StdP: 12.02

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.4	12.6	0.2	6.8	15.7	4.4	8.4	21.1	27.2	33.4	23.4	37.5	3.4	90	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	95.2	61.0	92.6	60.7	89.8	60.4	65.3	87.8	63.8	85.6	62.5	83.9	10.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.1	85.4	74.6	55.7	80.9	73.4	54.3	76.8	72.6	33.3	87.5	32.0	85.6	30.9	83.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.0	22.5	18.5	0.2	99.0	7.2	2.9	-5.0	101.1	-9.2	102.8	-13.3	104.4	-18.6	106.5		
			-0.1	67.7	7.1	3.0	-5.3	69.8	-9.4	71.6	-13.4	73.3	-18.6	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	30.0	33.7	42.7	51.6	60.8	69.9	75.4	75.0	66.6	55.9	42.9	34.7
	DBStd	16.98	8.64	7.98	7.57	6.51	5.20	5.17	4.88	3.99	5.23	5.87	6.81	7.13
	HDD50	2096	619	457	249	55	1	0	0	0	16	224	475	
	HDD65	5147	1084	876	692	403	148	15	1	42	284	662	939	
	CDD50	3332	0	1	22	102	337	597	787	776	497	199	13	1
	CDD65	909	0	0	0	1	19	162	324	311	89	3	0	0
	CDH74	11693	0	0	2	26	340	2127	3810	3856	1416	116	0	0
	CDH80	5133	0	0	0	2	48	846	1871	1885	474	7	0	0
Wind	WSAvg	7.1	6.4	6.9	7.9	7.4	7.2	7.6	8.3	7.9	7.0	6.1	6.0	6.0
Precipitation	PrecAvg	11.6	1.1	1.2	1.5	2.1	1.5	0.4	0.3	0.2	0.1	0.7	1.5	1.1
	PrecMax	15.9	2.5	2.1	3.7	4.2	3.1	2.0	1.3	0.9	0.7	2.3	5.0	2.6
	PrecMin	6.7	0.4	0.3	0.1	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.1	0.5	0.5	0.8	1.0	0.8	0.5	0.4	0.2	0.2	0.6	1.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	57.6	71.4	77.5	84.4	95.8	100.0	97.7	91.5	80.7	65.9	59.1
		MCWB	40.6	43.9	48.6	53.3	55.8	59.5	62.2	60.6	59.7	55.0	47.2	43.3
	2%	DB	48.5	53.2	65.5	72.0	80.9	91.3	96.4	94.7	88.1	77.2	62.2	52.8
		MCWB	38.1	41.0	45.8	50.7	55.2	59.9	61.9	60.5	58.7	52.9	45.6	40.9
	5%	DB	45.0	49.1	60.6	68.3	78.0	87.9	93.0	92.2	85.6	73.8	58.6	48.8
		MCWB	36.4	38.8	44.1	49.0	54.1	58.4	61.6	60.2	57.6	51.7	44.3	39.0
	10%	DB	42.1	45.7	56.6	64.9	74.8	85.0	89.7	89.7	82.4	70.1	54.7	45.3
		MCWB	35.1	37.2	42.3	47.9	53.3	57.9	61.4	60.1	56.4	50.5	43.3	37.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	45.3	49.8	55.3	59.6	64.6	67.2	67.4	64.5	57.8	50.7	45.0
		MCDB	49.2	55.6	66.4	73.8	77.3	86.9	91.5	89.6	85.6	74.5	60.1	54.7
	2%	WB	39.4	42.3	47.3	52.7	57.4	62.7	65.6	65.1	61.3	54.9	48.4	42.4
		MCDB	46.1	50.6	62.1	66.8	75.1	84.4	89.0	86.7	82.5	72.5	56.5	50.5
	5%	WB	37.3	39.9	45.4	50.9	55.8	61.0	64.1	63.5	59.6	53.3	46.4	40.0
		MCDB	44.0	47.0	58.1	63.8	72.8	82.8	86.6	84.7	80.3	69.2	54.3	47.2
	10%	WB	35.3	37.9	43.3	49.4	54.6	59.4	62.8	62.1	58.0	51.8	44.6	38.0
		MCDB	41.1	44.5	54.5	61.6	70.8	80.4	84.7	83.0	78.0	66.5	52.9	44.2
Mean Daily Temperature Range	5% DB	MDBR	17.5	18.1	20.7	21.4	24.3	27.9	27.3	29.4	30.1	25.4	19.9	17.9
		MCDBR	20.0	22.9	26.8	26.6	28.7	31.8	31.5	32.5	33.4	30.5	25.6	22.9
		MCWBR	12.3	13.7	13.6	11.8	11.1	11.3	10.6	11.9	14.1	14.2	13.5	13.7
	5% WB	MCDBR	17.7	19.0	23.1	22.6	26.2	29.1	28.6	29.1	31.2	27.5	20.0	20.0
	MCWBR	11.4	11.9	12.5	11.0	10.9	11.6	10.5	11.4	13.7	13.6	11.4	12.5	
Clear-Sky Solar Irradiance	taub	0.303	0.341	0.395	0.472	0.466	0.453	0.491	0.451	0.396	0.391	0.330	0.300	
	taud	2.326	2.217	2.072	1.901	1.956	2.013	1.928	2.034	2.166	2.136	2.300	2.361	
	Ebn at Noon	280	281	273	258	262	265	253	261	270	258	266	273	
	Edh at Noon	30	38	49	61	59	55	60	53	43	41	31	27	
All-Sky Solar Radiation	RadAvg	841	1136	1440	1699	2111	2514	2460	2276	1876	1310	873	710	
	RadStd	66	125	107	144	131	134	77	62	77	87	106	90	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+1.19	N/A	N/A	+2.12	+1.49	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.72	N/A	N/A	+1.78	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page